

1. Administrative & Study Identification

Full Study Title: A PHASE 3, RANDOMIZED, PLACEBO-CONTROLLED, OBSERVER-BLINDED TRIAL TO EVALUATE THE SAFETY OF A 6-VALENT OspA-BASED LYME DISEASE VACCINE (VLA15) IN HEALTHY CHILDREN 5 THROUGH 17 YEARS OF AGE **Short Title/Acronym:** Safety Study of a Vaccine to Help Protect Against Lyme Disease in Healthy Children **Protocol Number:** C4601012 **NCT Number:** NCT05634811 **Sponsor Name:** Pfizer (in collaboration with Valneva) **Investigational Product:** VLA15 (6-valent OspA-based Lyme disease vaccine) **Phase of Development:** Phase 3 **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To evaluate the safety and tolerability of the VLA15 Lyme disease vaccine in healthy children aged 5 through 17 years. **Secondary Objectives:** To monitor for unsolicited and solicited adverse events, severe reactions following vaccine administration, and assess the long-term safety profile over a 2-year period.

2.2 Background & Rationale Lyme disease is the most prevalent vector-borne disease in the Northern Hemisphere, caused by *Borrelia burgdorferi* and transmitted via tick bites. It occurs most frequently in children aged 5 to 17. Currently, there is no approved human vaccine for Lyme disease. VLA15 is an investigational multivalent protein subunit vaccine targeting the outer surface protein A (OspA). The goal of this Phase 3 trial is to provide robust safety evidence for the study vaccine in this pediatric age group before seeking regulatory approval.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled (Normal Saline) **Blinding:** Observer-blinded **Randomization:** Randomized (3:1 ratio of VLA15 to Placebo)

3.2 Planned Enrollment & Duration **Target \$N\$:** Approximately 3,500 participants **Number of Sites:** Multicenter (United States) **Estimated Study Start:** 12-Dec-2022 **Estimated Primary Completion:** 21-Jul-2025

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Healthy children, age 5 through 17 years old at the time of enrollment. 2. Participants and/or parent(s)/guardian willing and able to comply with all scheduled visits, study procedures, and lifestyle considerations. 3. Stable preexisting chronic medical conditions are permitted.

4.2 Exclusion Criteria 1. Any diagnosis of Lyme disease within the past 3 months, or history of Lyme arthritis, carditis, neuroborreliosis, or other disseminated Lyme disease regardless of when diagnosed. 2. Receipt of a previous vaccination for Lyme disease. 3. Known tick bite within the past 4 weeks, or any congenital or acquired immunodeficiency/immunosuppression.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: A total of 4 doses administered as a "shot" in the arm. The primary series consists of 3 doses administered at months 0, 2, and 6, followed by a booster dose at month 18 (approximately 1 year after the third dose). **Route of Administration:** Intramuscular injection **Duration of Treatment:** Approximately 2 years (24 months) in total.

5.2 Concomitant & Prohibited Therapy Permitted: Inhaled/nebulized, intra-articular, intrabursal, or topical corticosteroids. **Prohibited:** Receipt of systemic corticosteroids for ≥ 14 days within 28 days before study intervention; receipt of chronic systemic treatment with other known immunosuppressant medications, immunoglobulins, or radiotherapy within 6 months before study intervention.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	Screening/Baseline	Month 0 Informed Consent/Assent, Medical History, Eligibility check, Dose 1
2	Vaccination Visits	Months 2, 6, 18 Administration of Dose 2, Dose 3, and Dose 4 (Booster), respectively
3	Follow-Up	Post-Vaccination eDiary entries (documenting reactions in the week post-vaccination)
4	End of Study	Month 24 Final safety visit / follow-up, Exit Interview

(Note: Participants will have a total of 6 planned in-office visits [4 vaccination, 2 follow-up] and are contacted by phone 1 time every year to monitor their experience)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: The safety and tolerability profile of VLA15, measured by the proportion of pediatric participants experiencing solicited local/systemic adverse events, unsolicited adverse events, and serious adverse events (SAEs). **Measurement Method:** Patient/Parent daily eDiary entries collected for one week post-vaccination, supplemented by investigator review at scheduled visits and annual phone calls.

7.2 Secondary & Exploratory Endpoints - Incidence of severe reactions following any given vaccine dose. - Long-term safety evaluation across the 24-month study follow-up.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Patient/Parent eDiary completion to document localized and systemic reactions occurring in the week after vaccination, plus scheduled follow-up visits/phone calls. **Serious Adverse Events (SAEs):** Evaluated continuously; standard reporting timeline (typically 24 hours). Ad-hoc extra visits are mandated if a severe reaction occurs after a vaccine dose. **Data Monitoring Committee (DMC):** Independent Data Safety Monitoring Board (DSMB) review of cumulative safety data to ensure ongoing participant safety.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all randomized participants who received at least one dose of VLA15 or placebo). **Sample Size Justification:** $N=3,500$ (randomized 3:1) properly powered to capture a robust safety database and identify potential rare adverse events in a pediatric demographic. **Handling of Missing Data:** Standard multiple imputation (MI) or handling per standard operating procedures for vaccine safety trials.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical site monitoring, centralized data review, and regular DSMB safety assessments. **Audit Trail:** Secure electronic data capture (EDC) systems and eDiary technologies ensuring full audit trails and locking mechanisms for data clarification.

11. Study Identifiers & Governance

Primary ID: C4601012 **Registry Number:** NCT05634811 **Sponsor/Lead Organization:** Pfizer **Study Title:** Safety Study of a Vaccine to Help Protect Against Lyme Disease in Healthy Children **Official Regulatory Title:** A PHASE 3, RANDOMIZED, PLACEBO-CONTROLLED, OBSERVER-BLINDED TRIAL TO EVALUATE THE SAFETY OF A 6-VALENT OspA-BASED LYME DISEASE VACCINE (VLA15) IN HEALTHY CHILDREN 5 THROUGH 17 YEARS OF AGE

12. Clinical Framework (Lyme Disease Specific)

Primary Condition: Lyme Disease **Diagnosis Threshold:** N/A (Prophylactic Phase 3 Vaccine Safety Trial in healthy, uninfected individuals) **Medical Methodology:** Active Immunization (Recombinant Protein Subunit Vaccine) **Optimization Target:** Prevention of *Borrelia burgdorferi* infection and assessment of vaccine safety/tolerability

13. Study Procedures & Methodology

Management Pathway: Prophylactic vaccination schedule **Education Protocol (HCP):** Site initiation, observer-blinded protocol administration training, and AE evaluation **Education Protocol (Patient):** Parent/Guardian training on electronic diary (eDiary) tracking for post-vaccination reactions **Quality Audit Mechanism:** eDiary compliance tracking & independent safety monitoring

14. Observation & Follow-Up Schedule

Total Duration: Approximately 2 years (24 months) **Onsite Visit Cadence:** 6 planned visits in total (4 vaccination visits at Months 0, 2, 6, 18, and 2 follow-up visits) **Remote Monitoring Frequency:** 1 telephone contact every year during the study **Checklist Review Interval:** 7-day eDiary review immediately following each vaccination dose

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				